

S. M. H. Ghaseminejad

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Open to relocation across Europe, including France

SUMMARY

Senior Software Engineer with **8+ years** of experience building **high-performance, low-latency distributed systems** in **financial services** and telecommunications. Strong expertise in **modern C++ (C++14/17/20)**, **concurrent programming**, **real-time processing**, and **scalable backend systems**.

Primarily a **C++ engineer** with production experience in **Java enterprise backend development**, including **Spring Boot services**, **REST/SOAP integrations**, **Kafka-based event-driven systems**, and **card payment switch platforms** processing **high-volume financial traffic** (millions of transactions/day class systems).

Experienced in building **transaction-processing systems**, **event-driven architectures**, and **distributed financial infrastructure** with strong focus on **scalability**, **reliability**, **concurrency**, and **performance optimization**.

CORE SKILLS

Programming Languages: C++ (14/17/20), Java (17), Python, SQL

Backend & Distributed Systems: Multithreading, Concurrency, Distributed Systems, Low-Latency Systems, Real-Time Processing, High-Throughput Architectures

Java Backend: Spring Boot, REST APIs, SOAP Services, Enterprise Integration Systems

Messaging: Kafka, Event-Driven Architecture, Asynchronous Processing

Databases: PostgreSQL, MySQL, Redis

Tools: STL, Boost, OpenSSL, Qt, Git, Docker, CI/CD (GitLab)

Testing: Google Test, Google Mock, TDD

OS: Linux (RHEL, Ubuntu, CentOS)

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Financial Transaction & Payment Infrastructure

Informatics Service Corporation | 2020 - PRESENT

TEHRAN, IRAN

Contributing to the development of a high-throughput financial transaction platform processing real-time operations with strict latency, consistency, and reliability requirements

Key Contributions:

- Developed **high-performance backend modules** for a **card payment switch system** processing **millions of daily transactions**, operating under strict **low-latency (<100ms end-to-end transaction paths in typical flows)** requirements
- Contributed to **C++ and Java-based switch systems** supporting **transaction routing**, **financial workflows**, and **enterprise integration services** in production banking environments
- Developed **Spring Boot microservices** for **OTP authentication**, handling **high-concurrency request bursts (>1,000 requests/sec peak load scenarios)**
- Implemented **REST and SOAP-based integration services** for communication between card payment switch and external banking systems, supporting **real-time inquiry and transaction validation workflows**
- Built and integrated **cURL-based API clients** for external financial service connectivity and inquiry aggregation layers
- Designed and implemented a **Redis-based OTP authentication system**, improving authentication response time by **~60-70% vs database-based lookup design**
- Developed a **Kafka-based event-driven integration layer** handling **asynchronous financial event streams (transaction + operational events)** with **high-throughput, decoupled architecture**
- Implemented **Kafka producer/consumer pipelines** supporting **near real-time event propagation (sub-second delivery in typical flows)**
- Built a **concurrent transaction simulation framework** used for **stress testing payment workflows under multi-threaded high-load scenarios (thousands of simulated transactions per run)**
- Improved backend throughput by **~30-40%** through **multithreading optimization**, **lock contention reduction**, and **pipeline restructuring**
- Worked with **SQL-based transaction processing systems**, optimizing queries and integration flows in production banking environments
- Implemented secure communication components using **C++ and OpenSSL**, ensuring encrypted data exchange across distributed financial services
- Collaborated in **agile teams (QA, product owners, backend engineers)** delivering production-grade financial systems
- Contributed to **CI/CD pipelines**, automated testing, and release stability improvements

SIGNAL PROCESSING C++ DEVELOPER - Real-Time Distributed Data Processing Systems

RASAPARDAZ | 2017 - 2020

TEHRAN, IRAN

Worked on scalable, high-performance distributed systems processing real-time streaming data under strict latency constraints.

Key Contributions:

- Developed modular **C++ components** for **real-time distributed systems processing high-volume streaming data (telecom-scale workloads)**
- Designed scalable backend services handling **continuous data streams with strict latency constraints (sub-100ms processing targets in critical paths)**
- Implemented **high-performance networking modules using Boost.Asio**, supporting **multi-threaded data ingestion pipelines**
- Improved system throughput by **~25–35%** through **concurrency optimization, lock-free design improvements, and memory efficiency enhancements**
- Optimized data ingestion pipelines handling **high-throughput streaming workloads (thousands of events/sec class systems)**
- Enhanced system stability under heavy load by improving **thread scheduling, buffering strategies, and backpressure handling mechanisms**
- Participated in **system architecture design, code review, and performance tuning for production systems**

C++ DEVELOPER - VoIP & Embedded Communication Systems

SAFIR COMMUNICATION INDUSTRIES | 2016 - 2017

TEHRAN, IRAN

Tech Stack: C++11/14, Embedded Linux, Qt, SIP, TCP/IP

- Developed backend modules for **VoIP and SIP-based communication systems using C++**
- Implemented **multithreaded TCP/IP networking services** on Linux platforms for real-time communication
- Optimized system performance for **low-latency communication workloads (<50ms message handling paths in internal systems)**
- Worked on embedded Linux-based systems with focus on **reliability and deterministic behavior under constrained environments**

PROJECT HIGHLIGHTS

Card Payment Switch Platform

- Developed core modules of a **high-throughput payment switch system handling millions of transactions/day scale workloads**
- Worked on both **C++ and Java components** supporting transaction workflows and backend financial services
- Improved system stability and processing efficiency under strict latency and correctness constraints

Kafka-Based Financial Integration Platform

- Designed and implemented a **Kafka-based event-driven system handling real-time financial event streams**
- Developed **producer/consumer services** enabling asynchronous processing of transaction and operational events
- Supported **high-throughput messaging pipelines with sub-second event propagation in production flows**

Real-Time High-Throughput Data Processing System

- Worked on distributed systems processing **high-volume streaming data (telecom-scale, thousands of events/sec)**
- Optimized **C++ processing pipelines**, improving throughput by **~30%** through **multithreading and pipeline tuning**

PROFESSIONAL AFFILIATIONS

- **Member, Institute of Electrical and Electronics Engineers (IEEE)**
- **Reviewer of journal of machine learning and cybernetics (Springer)**

LANGUAGE SKILLS

- **English:** Professional Working Proficiency, **Persian:** Native

EDUCATION

2007-2010 **Master of Science in Electrical Engineering** | [Guilan University, Rasht, Iran](#)

2002-2007 **Bachelor of Biomedical Engineering (Bioelectric)** | [Amirkabir University of Technology \(AUT\), Tehran, Iran](#)

INTERESTS

- **Capital Markets and Trading Systems**
- **Financial Technology Platforms**
- **Distributed Backend Architectures**
- **High-Performance Concurrent Systems**